Supervyse OPF Intel Tech Pack PFR User Guide

Revision 1.0

06/2023

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Revision History

Revision Number	Date	Author	Reviewer	Summary
1.0	2023-06-14	Allen Liu	Edwin Liu	Initial Release

1 Introduction

- Intel® PFR is designed to support the NIST requirements of SP 800-193 document. The goal of Intel® PFR technology is to provide resiliency through Protecting platform assets, Detecting corrupted firmware and malicious or erroneous behavior, and recovering the platform to a known good state.
- NIST lists the following three guiding principles to support the resiliency of platforms against potentially destructive attacks:
    - Protection: Mechanisms for ensuring that Platform Firmware code and critical data remain in a state of integrity and are protected from corruption, such as the process for ensuring the authenticity and integrity of firmware updates.
    - Detection: Mechanisms for detecting when Platform Firmware code and critical data have been corrupted.
    - Recovery: Mechanisms for restoring Platform Firmware code and critical data to a state of integrity if any such firmware code or critical data are detected to have been corrupted or forced to recover through an authorized mechanism. Recovery is limited to the ability to recover firmware code and critical data.
- In addition, NIST SP 800-193 document provides guidance on meeting those requirements via three main functions of a Platform Root of Trust:
    - Root of Trust for Update (RTU) which is responsible for authenticating firmware updates and critical data changes to support platform protection capabilities, this includes signature verification of firmware updates as well as rollback protections during update.
    - Root of Trust for Detection (RTD) which is responsible for firmware and critical data corruption detection capabilities.
    - Root of Trust for Recovery (RTRec) which is responsible for recovery of firmware and critical data when corruption is detected or when instructed by an administrator.
- This chapter will introduce how to generate a signed image/capsule for AST2600.
- A PFR image layout should be divided into major sections Active, Recovery, and Staging.
- Active area. This is used by the system during normal boot. Active section is described by the Platform Firmware Manifest (PFM).
- Recovery area. This is used to store the recovery image. This area is read-only during normal system boot. In case of corruption in the active area, content from the recovery image is restored into it.
- Staging area. This holds the candidate copy for an update of the recovery image. Staging area can be written during the normal boot mode.

2 Integrate PFR package into Insyde OPF SDK

2.1 Steps

- Download Insyde Supervyse OPF source code (RV23.05 or later) and put pfr_package.tar.gz to the OPF SDK root path
- Decompress pfr_package.tar.gz tarball to the SDK root path.
- Create recipes-pfr and pfr_information into meta-insyde/meta-intel and SPF5X/web/optional folder
    - mkdir -p meta-insyde/meta-intel/recipes-pfr
- Copy PFR3 package to meta-insyde/meta-intel
    - cp -R pfr_package/PFR3/. meta-insyde/meta-intel/recipes-pfr/

3 Building Code

Please refer to Supervyse OPF Introduction to Build Environment to set up the environment.

3.1 ast2600evb platform

- Select flash size
    - meta-insyde/meta-intel/meta-evb/conf/machine/include/selected_flashsize.inc

```
FLASH_SIZE = "262144"
```

- Specify platform

. setup intel-ast2600-evb

- Enable "intel-pfr" flag
    - build/conf/local.conf

```
IMAGE_FSTYPES += "intel-pfr"
```

- Actual build

bitbake intel-platforms

3.2 Intel Eagle Stream platform

- Select flash size
    - meta-insyde/meta-intel/meta-archercity/conf/machine/include/selected_flashsize.inc

```
FLASH_SIZE = "262144"
```

- Specify platform

. setup intel-bhs-insyde

- Enable "intel-pfr" flag
    - build/conf/local.conf

```
IMAGE_FSTYPES += "intel-pfr"
```

- Actual build

bitbake intel-platforms

3.3 PFR image

- The bmc pfr full image and bmc capsule files will be generated in build/tmp/deploy/images/${platform}/pfr_images.
    - image-mtd-pfr.bin: PFR full image for BMC flash
    - bmc_signed_cap.bin: BMC capsule for updating active/recovery

4 BMC PFR image layout

- The BMC PFR image consists of one **256MB** binary which primarily includes **active**, **recovery**, **staging** region, and **Platform Firmware Manifest (PFM)**

(describing active platform firmware) and the partition layout of the BMC PFR image is shown below

Contents	Start Address	Size (Hex)	Static/Dynamic
uboot	0x0	0x0010_0000 (1024KB)	static
fit-image-a	0x0010_0000	0x0368_0000 (56320KB)	static
sofs	0x0378_0000	0x0020_0000 (2048KB)	Dynamic
rwfs	0x0398_0000	0x005e_0000 (6016KB)	Dynamic
cerificate	0x03f6_0000	0x0001_0000 (64KB)	Dynamic
sdr	0x03f7_0000	0x0001_0000 (64KB)	Dynamic
u-boot-env	0x03f8_0000	0x0002_0000 (128KB)	Dynamic
reserved1	0x03fa_0000	0x0a4_0000(10496KB)	Dynamic
pfm	0x049e_0000	0x0002_0000 (128KB)	static
rc-image	0x04a0_0000	0x0400_0000(64MB)	static
image-stg	0x08a0_0000	0x0400_0000 (64MB)	Dynamic
pch-stg	0x0ca0_0000	0x0100_0000 (16MB)	Dynamic
pfr-stg	0x0da0_0000	0x0040_0000 (4MB)	Dynamic
afm-active	0x0de0_0000	0x0002_0000 (128KB)	Dynamic
afm-recovery	0x0de2_0000	0x0002_0000 (128KB)	Dynamic
reserved2	0x0de4_0000	0x000c_0000 (768KB)	Dynamic
cpld-gold	0x0df0_0000	0x0010_0000 (1024KB)	Dynamic
reserved3	0x0e00_0000	0x0200_0000 (2048KB)	Dynamic

5 Firmware update

5.1 via Redfish

- Export BMC IP

```
export bmc=192.168.xx.xx
```

- Export Token

```
export token=`curl -k -H "Content-Type: application/json" -X POST https://${bmc}/login -d \
'{"username" :  "root", "password" :  "0penBmc"}' | grep token | awk '{print $2;}' | tr -d ""`
```

- The below properties are used to indicate PFR intent.

The default will be using the Active method if non of the properties are included.
PFR Update Intent option

PFR Update Intent option

UpdateRegion	"Active"	Update Active region
	"StandbySpare"	Update Recovery region
@Redfish.OperationApplyTime	"Immediate"	Directly update the process
	"OnReset"	Manually power cycle the host to continue the update process
ClearConfig	false	No action
	true	Restore to default setting after Updating Firmware

Simple Update

- use the image on the TFTP server to update.

```
curl -k -H "X-Auth-Token: $token" -X POST https://${bmc}/redfish/v1/UpdateService/Actions/UpdateService.SimpleUpdate -d
'{"ImageURI":"tftp://${tftp_server_IP}/${image}", "UpdateRegion":"Active", "@Redfish.OperationApplyTime":"Immediate",
"ClearConfig" : false}'
```

Multipart Update

- Use the file on local to update.

```
curl -k -H "X-Auth-Token: $token" -X POST -H "Content-Type:multipart/form-data" \
-F 'UpdateParameters={"UpdateRegion":"StandbySpare", "@Redfish.OperationApplyTime":"OnReset", "ClearConfig" :
true};type=application/json' \
-F 'UpdateFile=@/file/path/bmc_signed_cap.bin;type=application/octet-stream' \
https://${bmc}/redfish/v1/UpdateService/multipartUpdate
```

5.2 via Web

- Go to Operations -> Firmware Page
- File Source

- Workstation
  - Add file button
    - Select update capsule file which is BMC/BIOS/CPLD image by local side and then click **Start Update** button to update

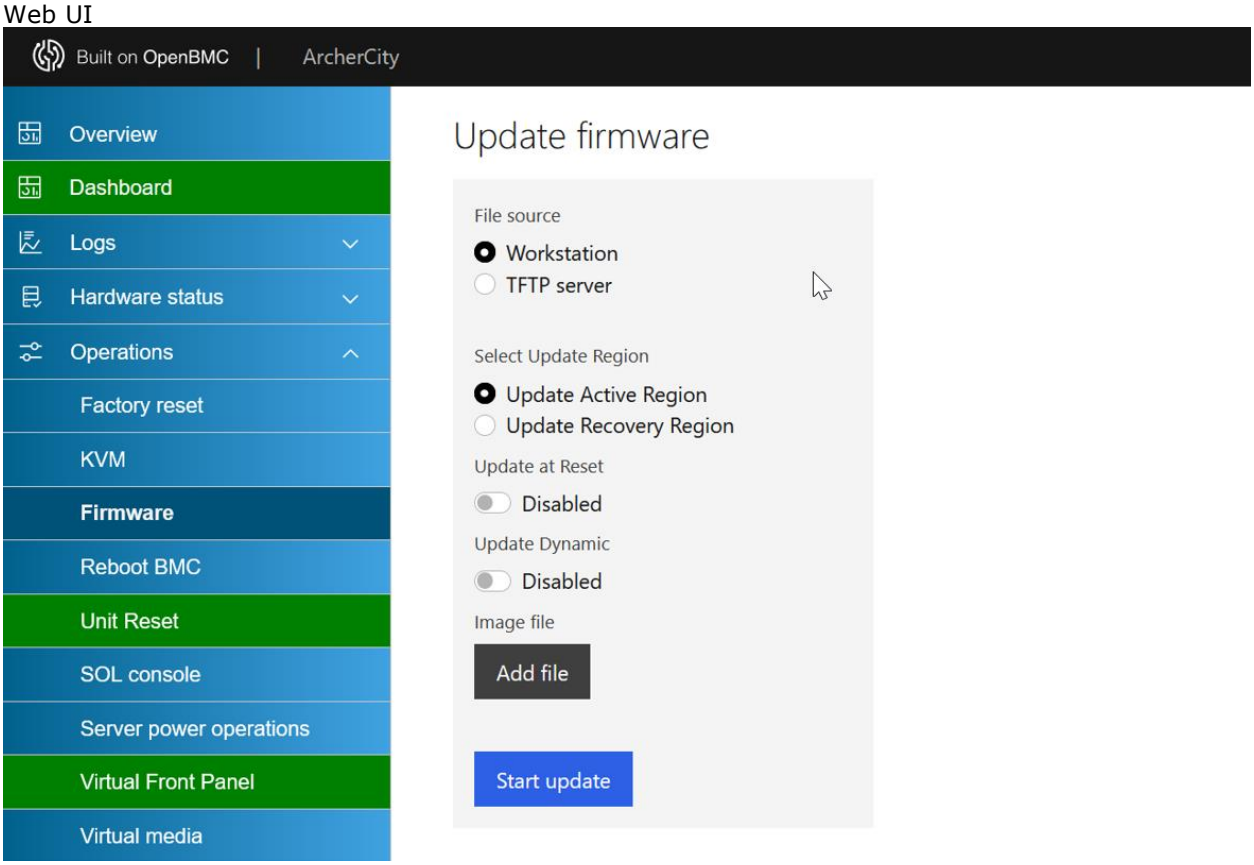
- TFTP server
  - file address:
    - ex: 192.168.22.153/bmc_signed_cap.bin

- There are 3 options added on Firmware update page.
  - Select Update Region
    - Update Active Region
    - Update Recovery Region

- Update at Reset
  - If this option is checked, the user needs to manually power cycle the host to continue the update process.

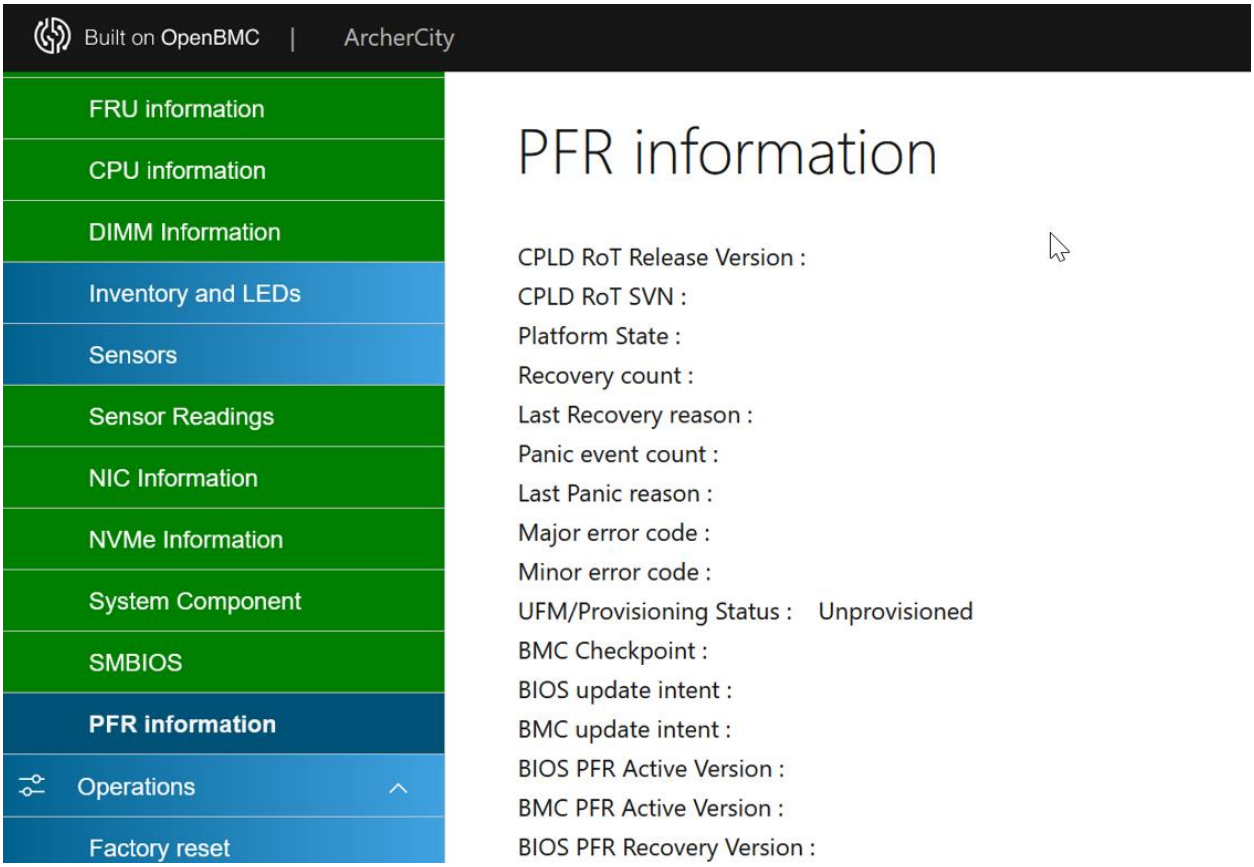
- Update Dynamic
  - Restore to default setting after updating firmware

- Web UI



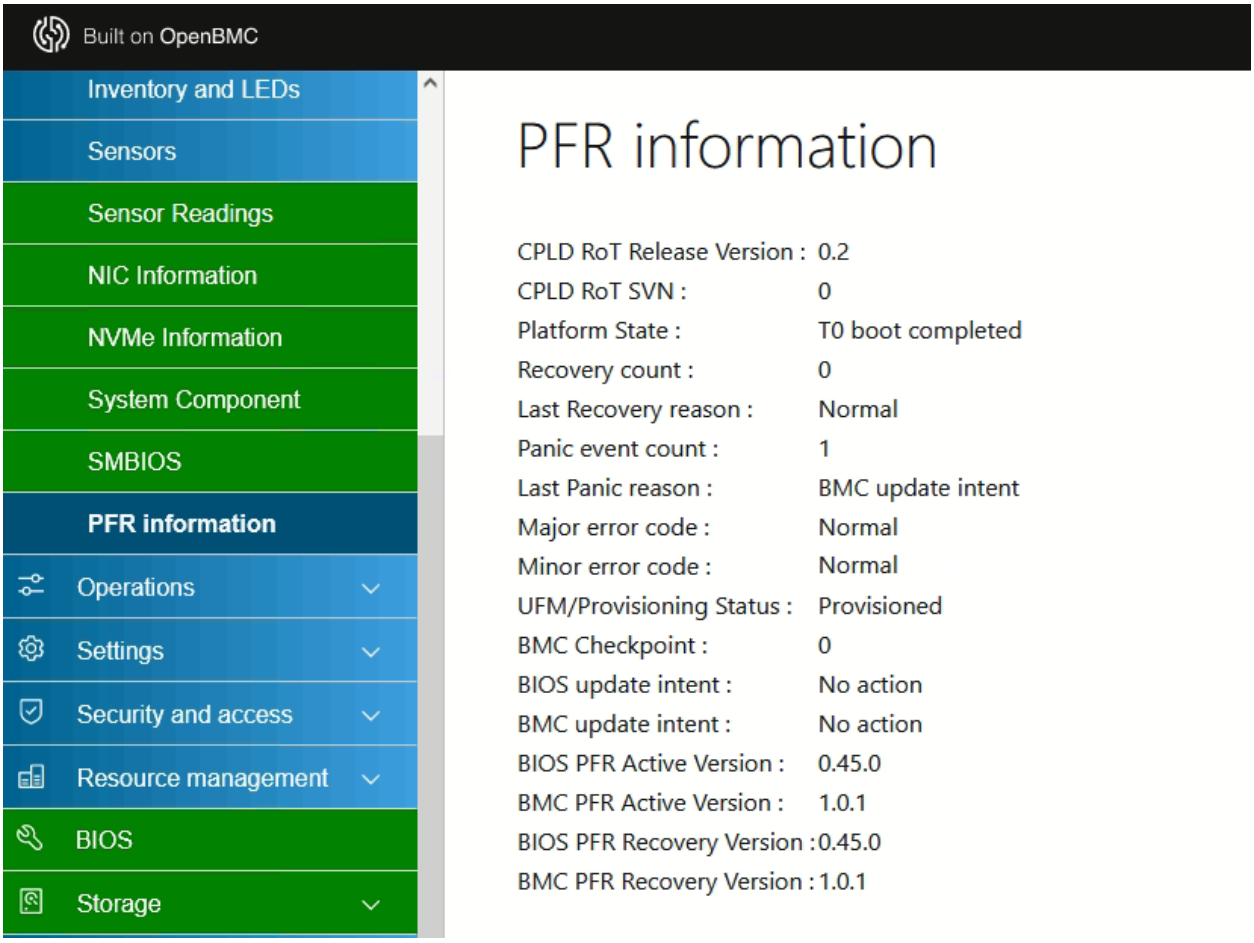
6 PFR Information via Web

- Log in to the Web GUI and press "Hardware status-> PFR information" page to check the PFR information
    - PFR Unprovisioned



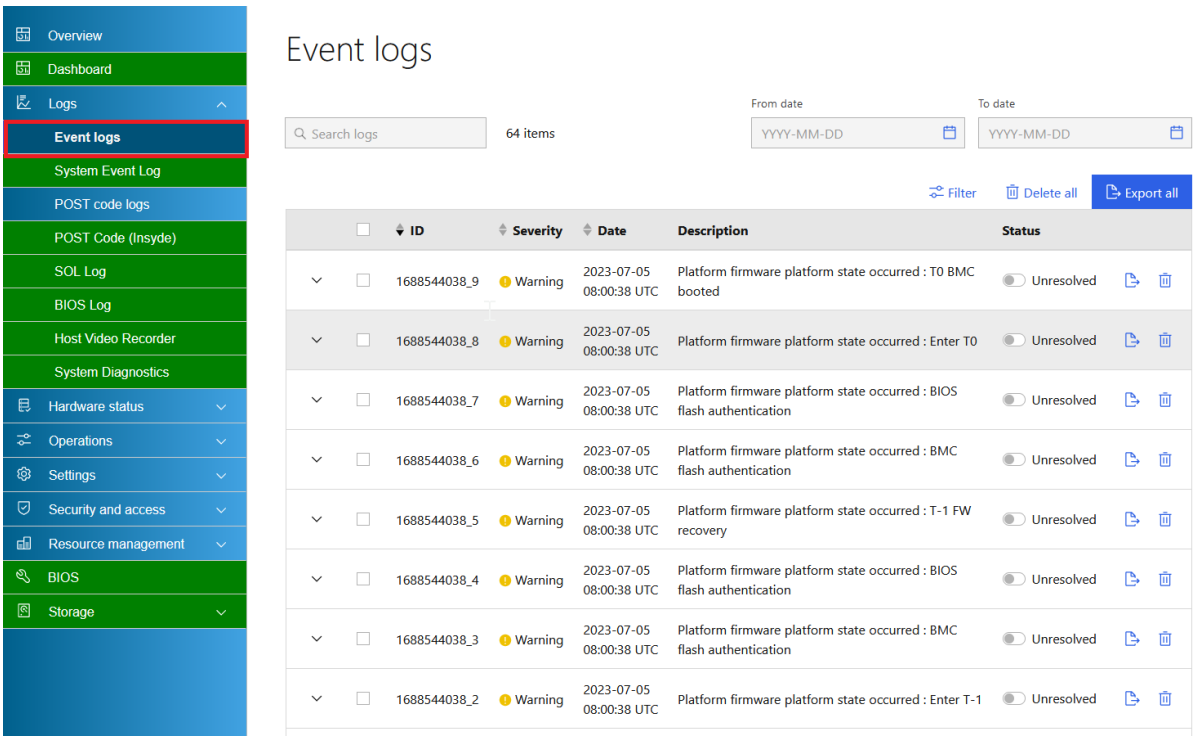
    - PFR Provisioned

For a detailed description of each item, you can refer to Appendix:Host Mailbox Register File



7 PFR Event Logs via Web

- The CPLD host mailbox register holds several pieces of information, such as panic reason, recovery reason, platform state, major and minor errors. Some of the information may be useful and should be logged into Event Logs. Users can login to the web GUI and press "Logs -> Event Logs" to check the PFR event logs.



8 How to provision/unprovision platform

- In UEFI internal shell, use pfrconfig tool to provision/unprovision PFR or get PFR provision state
    - provision

```
pfrconfig -p
```

```
UEFI Interactive Shell v2.2
EDK II
UEFI v2.80 (INSYDE Corp., 0x72161010)
map: No mapping found.
Press ESC in 1 seconds to skip startup.nsh or any other key to continue.
Shell> pfrconfig -p
INFO: Log Filename -> PfrConfig-p-20230101-000500.Log
ERROR: Log File Create error
WARNING: Log File -> Not generated

PfrConfig - Shell Command, Version 5.01, 04152021
Copyright (C) 2019-2021 Intel Corporation

BIOS ID: EGLSTRM.05.45.16.1010
Processor: CPUID 0x000806F6
INFO: PFR State Information {
      PFR State: 0xA100
                    PFR Supported? Yes
                   Number of CPLD: 0x01
                     Flash update? Not Allowed outside SMM
                       PFR Locked? No
                  PFR Provisioned? No
              PIT Level-1 Enabled? No
              PIT Level-2 Enabled? No
      PFR Max Function# (0-based): 0x1C
      PFR Region: 0x0E, Base Address 0x007F0000, Size 0x02810000 bytes
          Staging Area: Base Address 0x007F0000, Size 0x01400000 bytes
         Recovery Area: Base Address 0x01BF0000, Size 0x01400000 bytes
              PFM Area: Base Address 0x02FF0000, Size 0x00010000 bytes
INFO: PFR State Information } Success
INFO: Perform PFR Provision {
INFO: Perform PFR Provision } Success

INFO: Log File -> Not generated
==> Remove AC power from the system,
    Wait for at least 10 seconds,
    Reconnect AC power to the system to continue....
```

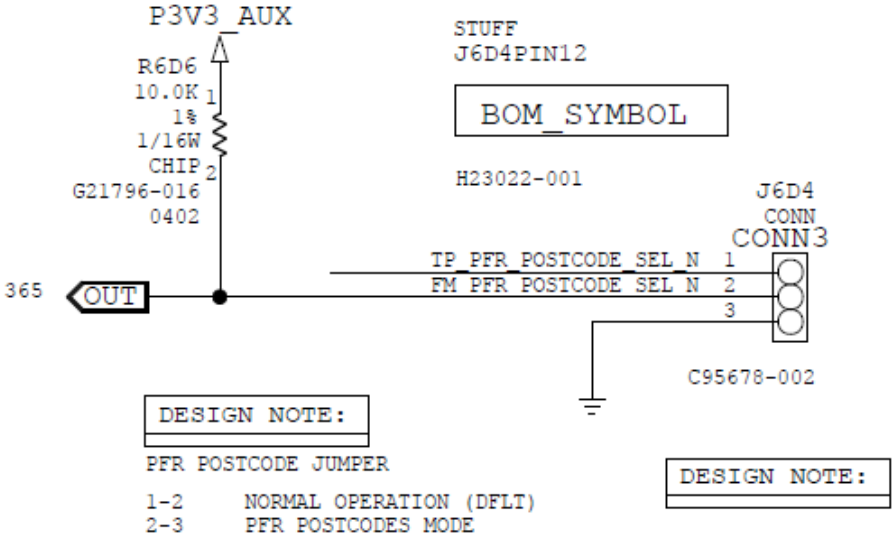
- Please confirm if the pfrconfig tool is correctly configured for BMC's three offset.

```
PFM offset      : 0x49E0000
Recovery offset : 0x4A00000
Staging offset  : 0x8A00000
```

9 Appendix

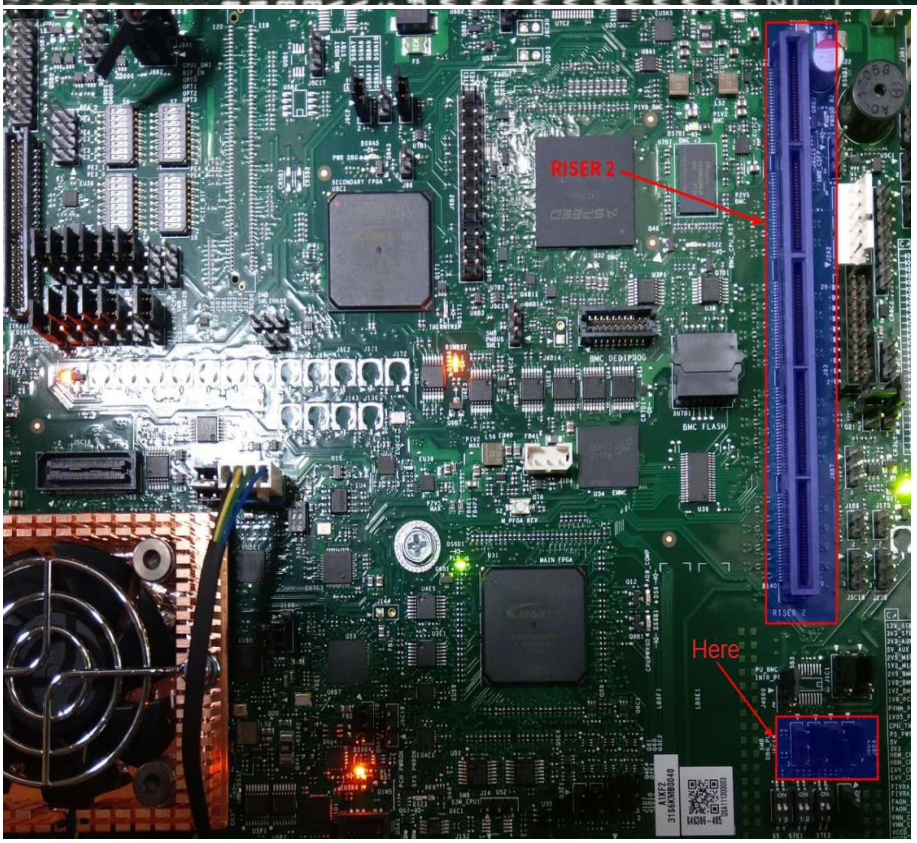
9.1 How to debug CPLD platform status

- You can change jumper J6D4 to pin 2-3 for PFR postcode mode, and CPLD host mailbox register 03h(platform state) will display to POSTCODE LED.



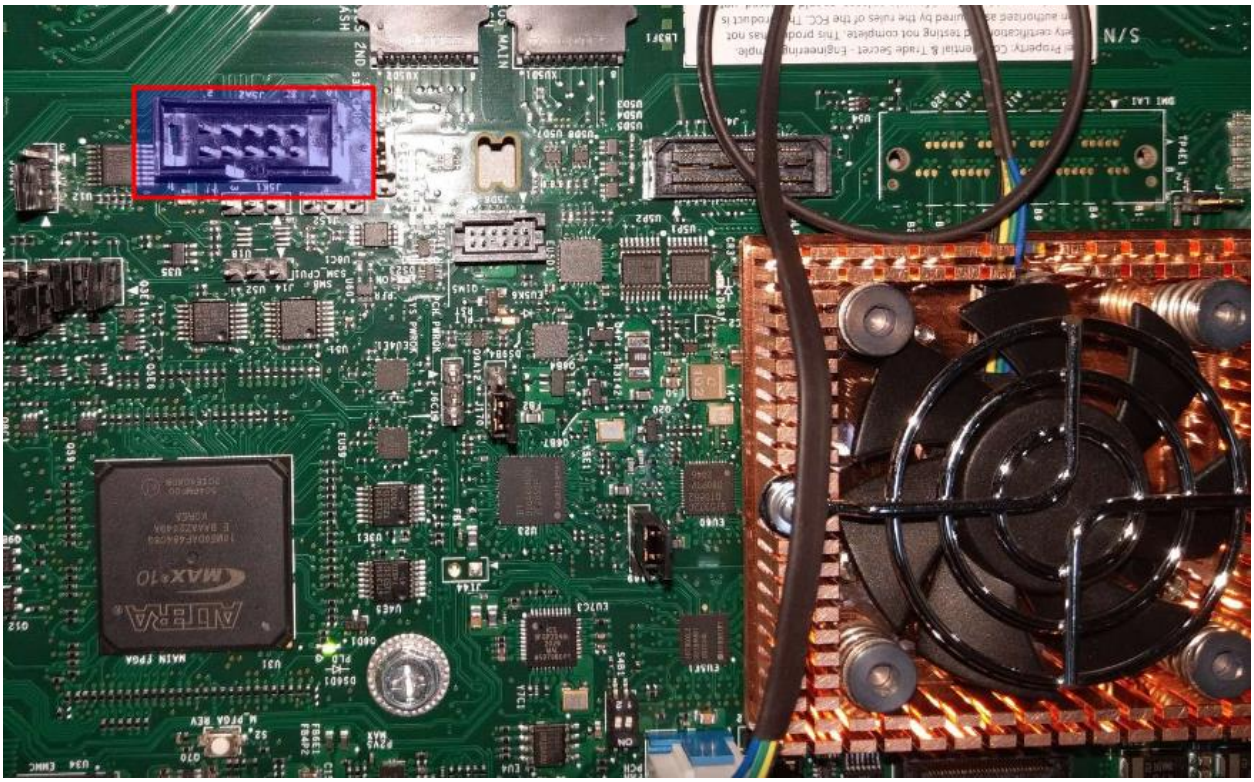
- Adjust jumper J6D4 and connect pin 2 and ping 3
- The J6D4 nears riser 2 slot

LB5E1
K4C386-405
QSA111300003
J4000
PU_BMC
INTR_P
U5B3
RISER 2
S5
S7E1
S7E2
J1C1
J8A2
J6D5
J6D4
J6D6
J6E1
J5C18
J130


RISER 2
Here

- Use System Console Debug Script?
    - Check Intel documentation Content ID: 611974 chapter 7 to setup Intel Quartus Prime software packages, and launch System Console as below shows:
        - Altera USB blaster

- Altera USB blaster connect to J5A2, which nears ALTERA chip

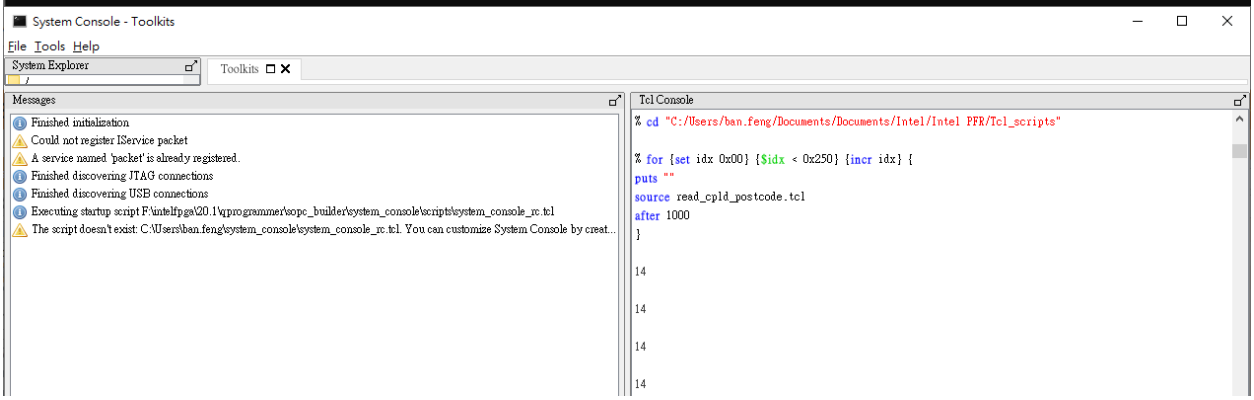


- launch System Console

```
F:\intelFPGA\20.1\qprogrammer\bin64>jtagconfig
1) USB-BlasterII [USB-1]
  031830DD   10M16S(A|C)
  031810DD   10M02SC


F:\intelFPGA\20.1\qprogrammer\bin64>jtagconfig --getparam 1 JtagClock
6M

F:\intelFPGA\20.1\qprogrammer\bin64>..\sopc_builder\bin\system-console.exe
```



9.2 How to use your own key to generate a PFR CPLD update capsule

- Download Intel PFR CPLD source from Intel RDC
    - For EagleStream platform: Refer to 633986_Intel®PFR_FPGA_Source_EagleStream _Rev1p7
    - With Intel® Quartus® Prime Pro Edition Software 20.1 to compile. Intel® Quartus® Prime Pro Edition Software 20.1 can be downloaded from https://fpgasoftware.intel.com
- After finishing compiling, it will generate pfr_cfm1_auto.rpd file from Quartus. In quartus_project folder of Intel PFR CPLD source, it provides blocksign, config xml, and keys related file to generate capsule file.

```
blocksign_tool
├── blocksign
├── blocksign-20200513-v616213.zip
├── config_xmls
│   ├── afm_key_cancellation_secp384.xml
│   ├── cpld_update_capsule_secp384r1_svn1.xml
│   ├── cpld_update_capsule_secp384r1_svn2.xml
│   ├── cpld_update_capsule_secp384r1.xml
│   ├── cpld_update_capsule_without_svn.xml
│   ├── cpld_update_capsule.xml
│   ├── decommission_capsule_secp384r1.xml
│   ├── decommission_capsule.xml
│   ├── pch_fv_seamless_capsule_key_cancellation_secp384r1.xml
│   ├── pch_pfm_key_cancellation_secp256.xml
│   ├── pch_pfm_key_cancellation_secp384r1.xml
│   ├── pch_pfm_secp384r1.xml
│   ├── pch_pfm.xml
│   ├── pch_update_capsule_secp384r1.xml
│   ├── pch_update_capsule.xml
│   ├── template_256.xml
│   └── template_secp384r1.xml
├── keys
│   ├── secp256r1-keys
│   │   ├── csk_prv.pem
│   │   ├── csk_pub.pem
│   │   ├── root_key_private.pem
│   │   └── root_key_public.pem
│   └── secp384r1-keys
│       ├── csk384_prv.pem
│       ├── csk384_pub.pem
│       ├── rk384_prv.pem
│       └── rk384_pub.pem
├── README.pdf
├── README.txt
```

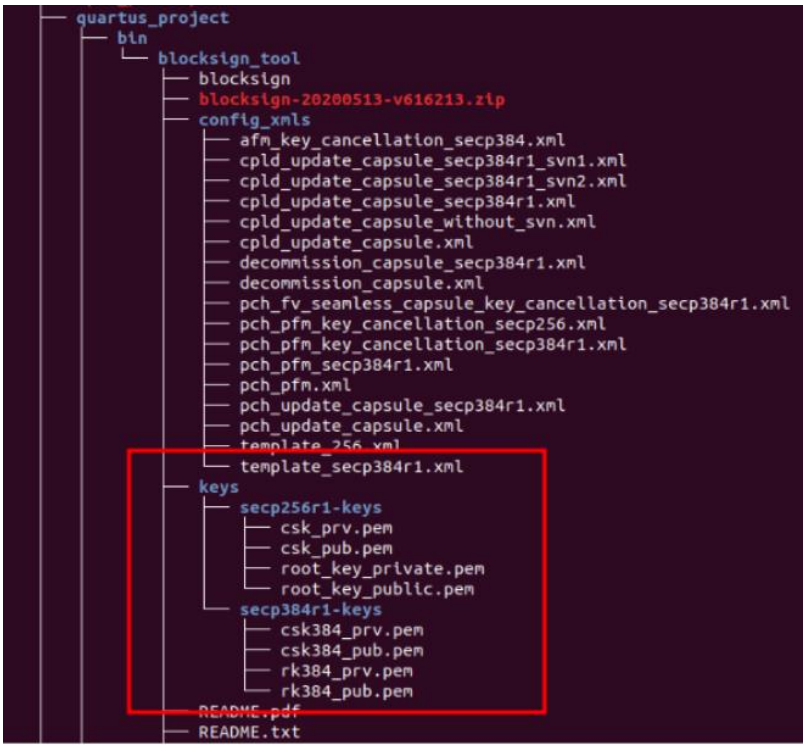
- How to generate Root Keys and Code Signing Keys
    - Refer to 611974_Intel® Platform Firmware Resilience (Intel® PFR) Tools Guide_Rev2p1.pdf chapter 2.2.3 Generate a Private or Public Key Pair with secp256r1 Curve.
- Please use your own key to override quartus_project/bin/blocksign_tool/keys
    - Generate scep256 cpld update capsule

```
# bin/blocksign_tool/source/blocksign -c bin/blocksign_tool/config_xmls/cpld_update_capsule.xml -o
output_files/cpld_update_capsule.bin output_files/pfr_cfm1_auto.rpd
```

- Generate scep384 cpld update capsule

```
# bin/blocksign_tool/source/blocksign -c bin/blocksign_tool/config_xmls/cpld_update_capsule_secp384r1.xml -o
output_files/cpld_update_capsule_secp384.bin output_files/pfr_cfm1_auto.rpd
```

- If your key name is different from default, you should also change name of key field in xml file

- 

- In addition, you need to generate the BMC/BIOS update capsule with your own key, which was generated in the previous step, and place your own key in the Intel pfrconfig tools folder.When you use pfrconfig -p command to do provision setting in Host UEFI, it will transfer root key hash to CPLD. CPLD will store it in UFM and use it to verify CPLD/BMC/BIOS update capsule.

- For the details, you can refer to 606686_Intel® PFR Architecture and Hardware Design Guide_Rev2p0.pdf chapter 2.3 Authentication of Data Structures by CPLD RoT

9.3 Host Mailbox Register File

Address	Name	Description
00h	CPLD RoT Static Identifier	Constant value of DEh
01h	CPLD RoT Release Version	Release version of FPGA RoT design, read-only for external agents. To be incremented with every release.
02h	CPLD RoT SVN	SVN of FPGA RoT design, read-only for external agents
03h	Platform State	Platform state, the following states should be enumerated:

		• 01h: CPLD NIOS II processor waiting to start • 02h: CPLD NIOS II processor started • 03h: Enter T-1 • 04h: T-1 Reserved • 05h: T-1 Reserved • 06h: BMC flash authentication • 07h: PCH flash authentication • 08h: Lockdown due to authentication failure • 09h: Enter T0 • 0Ah: T0 BMC booted • 0Bh: T0 ME booted • 0Ch: T0 ACM booted • 0Dh: T0 BIOS booted • 0Eh: T0 boot completed • 10h: PCH FW update • 11h: BMC FW update • 12h: CPLD update • 13h: CPLD update in recovery mode • 14h: PCH/CPU firmware volume update • 15h: PCH/CPU firmware volume update completed • 16-3Fh: Reserved • 40h: T-1 FW recovery • 41h: T-1 forced active FW recovery • 42h: WDT timeout recovery • 43h: CPLD recovery in recovery mode • 44h: Lockdown due to PIT L1 • 45h: PIT L2 FW sealed • 46h: Lockdown due to PIT L2 PCH hash mismatch • 47h: Lockdown due to PIT L2 BMC hash mismatch • 48h-FFh: Reserved Read-only for external agents
04h	Recovery count	Numbers of FW Recovery actions since last AC-on, read-only for external agents
05h	Last Recovery reason	Value indicating last FW Recovery reason, following states should be enumerated: • 01h: PCH active failure • 02h: PCH recovery failure • 03h: ME launch failure • 04h: ACM launch failure • 05h: IBB launch failure • 06h: OBB launch failure • 07h: BMC active failure

		• 08h: BMC recovery failure • 09h: BMC launch failure • 0Ah: CPLD WDT expired • 0Bh-FFh: Reserved Read-only for external agents
06h	Panic event count	Number of Panic events (T0 -> T-1 transitions) since last AC-on, read-only for external agents
07h	Last Panic reason	Value indicating last Panic reason, read-only for external agents • 01h: PCH update intent • 02h: BMC update intent • 03h: BMC reset detected • 04h: BMC WDT expired • 05h: ME WDT expired • 06h: ACM WDT expired • 07h: IBB WDT expired • 08h: OBB WDT expired • 09h: ACM/IBB/OBB signaled authentication failure • 0Ah-FFh: Reserved
08h	Major error code	Value indicating error code observed on the system • 01h: BMC authentication failure • 02h: PCH/CPU authentication failure • 03h: in-band update failure • 04h: out-of-band update failure • 05h: AFM update failure • 06h-FFh: Reserved • Some error types are tracked by recovery/panic events and omitted here
09h	Minor error code	Value indicating error code observed on the system BMC or PCH/CPU authentication failure • 01h: Active region authentication failure • 02h: Recovery region authentication failure • 03h: Active and Recovery region authentication failure • 04h: Active, Recovery and Staging region authentication failure FW/CPLD update failure • 01h: Invalid update intent • 02h: Update capsule has

		invalid SVN • 03h: CPLD capsule failed authentication • 04h: Exceeded maximum failed update attempts • 05h: Active firmware update is not allowed because the recovery region failed authentication in T-1 06h: FW update capsule failed authentication before being promoted to recovery region
0Ah	UFM/Provisioning Status	Status register for UFM provisioning and access commands; read-only • Bit[0]: Command Busy • Bit[1]: Command Done • Bit[2]: Command Error • Bit[3]: Reserved • Bit[4]: UFM locked • Bit[5]: UFM Provisioned (Root key + Region Offsets), set if the provision status bit in UFM is configured • Bit[6]: 1- PIT Level-1 enabled • Bit[7]: PIT Level-2 has been completed successfully
0Bh	UFM/Provisioning Command	Command register for UFM provisioning/access commands; read/write allowed from CPU/BMC. All non read commands are dropped (with Command Error bit set in the status register) after UFM is locked. Following single byte commands are accepted: • 00h: Erase current (not-locked) provisioning, invalid when the UFM is locked • 01h: Provision Root Key, invalid when the UFM is locked • 02h: Provision PIT Key, invalid when the UFM is locked • 03h: Reserved • 04h: Reserved • 05h: Provision PCH/CPU offsets (Active/PFM, Recovery, Staging), invalid when the UFM is locked • 06h: Provision BMC offsets (Active/PFM, Recovery, Staging), invalid

		when the UFM is locked • 07h: End of Provisioning / Lock UFM, invalid when the UFM is locked • 08h: Read Root Key • 09h-0Bh: Reserved • 0Ch: Read PCH/CPU offsets • 0Dh: Read BMC offsets • 0Eh: Reconfigure CPLD • 0Fh: Reserved • 10h: Enable PIT Level 1 protection • 11h: Enable PIT Level 2 protection • 12h: Enable Device Attestation Requests • 13h: Read DeviceID public key • 14h: Disable Device Attestation Requests • 15h-FFh: Reserved
0Ch	UFM Command Trigger	Trigger register for the command set in the previous offset. • Bit[0]: execute command specified at "UFM/Provisioning Command" register Note: Bit[0] will get automatically cleared when CPLD RoT consumes the command. • Bit[1]: flush Write FIFO • Bit[2]: flush Read FIFO Note: FIFO flush bits are orthogonal from execute command bit and should not be set together; such combination should result in Command Error bit set in status register.
0Dh	UFM Write FIFO	Ingress byte for Write First In First Out (FIFO); depth of the FIFO is implementation specific (for example, 16 Bytes); Writes push new byte to FIFO; reads return last pushed byte without affecting the FIFO
0Eh	UFM Read FIFO	Egress byte for Read FIFO; depth of the FIFO is implementation specific (for

		example, 16 Bytes); All writes are dropped; reads pop bytes from FIFO
0Fh	BMC Checkpoint	Status of BMC, write from BMC allowed until BMC signals boot complete, write from BMC re-allowed on BMC reset Execution Checkpoints • 0x01: Started execution block • 0x02: Next execution block authentication pass • 0x03: Next execution block authentication fail • 0x04: Exiting Platform manufacturer authority • 0x05: Starting external execution block • 0x06: Returned from external execution block • 0x07: Pausing execution block • 0x08: Resumed execution block • 0x09: Completing execution block • 0x0A: Entered management mode • 0x0B: Leaving management mode Miscellaneous Execution Checkpoints • 0x80: Host: Ready To Boot OS • 0x81: Host: Exit Boot Services • 0x82: Reset Host • 0x83: Reset ME • 0x84: Reset BMC • For Intel® PFR 3.0, to accommodate utilizing MCTP over the BMC SMBus, this address is reserved for MCTP
12h	PCH update intent	Update intent from the CPU FW, read/write allowed from CPU • 0x01: PCH active • 0x02: PCH recovery • 0x04: CPLD active • 0x08: BMC active • 0x10: BMC recovery • 0x20: CPLD recovery • 0x40: Update dynamic • 0x80: Update at reset Above update intent can be

		used as combination. Typical ones: • 0x03: PCH active and recovery update • 0x83: PCH active and recovery update at reset • 0x41: PCH active and dynamic update • 0xC1: PCH active and dynamic update at reset • 0x24: CPLD active and recovery update • 0x18: BMC active and recovery update • 0x09: PCH active and BMC active update • 0x12: PCH recovery and BMC recovery update • 0x1B: PCH FW and BMC FW update (0x03 + 0x18) • 0x3F: PCH, BMC, CPLD active and recovery update (0x1B + 0x24) • 0x48: BMC active and dynamic update • 0x37: BMC required platform reset update (BMC recovery, CLD active and recovery, PCH active and recovery update)
13h	BMC update intent	Update intent from BMC FW, read/write allowed from BMC. Refer to PCH update intent setting
14h	PCH PFM Active SVN	Set by FPGA RoT, read-only for CPU/BMC
15h	PCH PFM Active Major Version	Set by FPGA RoT, read-only for CPU/BMC
16h	PCH PFM Active Minor Version	Set by FPGA RoT, read-only for CPU/BMC
17h	BMC PFR Active SVN	Set by FPGA RoT, read-only for CPU/BMC
18h	BMC PFM Active Major Version	Set by FPGA RoT, read-only for CPU/BMC
19h	BMC PFM Active Minor Version	Set by FPGA RoT, read-only for CPU/BMC
1Ah	PCH PFR Recovery SVN	Set by FPGA RoT, read-only for CPU/BMC

1Bh	PCH PFM Recovery Major Version	Set by FPGA RoT, read-only for CPU/BMC
1Ch	PCH PFM Recovery Minor Version	Set by FPGA RoT, read-only for CPU/BMC
1Dh	BMC PFR Recovery SVN	Set by FPGA RoT, read-only for CPU/BMC
1Eh	BMC PFM Recovery Major Version	Set by FPGA RoT, read-only for CPU/BMC
1Fh	BMC PFM Recovery Minor Version	Set by FPGA RoT, read-only for CPU/BMC
60h	BMC Checkpoint (Intel® PFR 3.0)	Status of BMC, write from BMC allowed only until BMC signals boot complete, write from BMC re-allowed on BMC reset.
62h	BMC Update Intent 2	Update intent from BMC FW, read/write allowed from BMC • Bit[0]: Seamless update intent • Bit[1]: AFM active • Bit[2]: AFM recovery
63h	UFM and MM[9] SBMus ownership register	Ownership request details: • Bit[1:0] UFM ownership ID — 00: Indicates that UFM ownership is free — 01: Indicates that UFM ownership is with PCH/Host [CPU0] — 11: Indicates that UFM ownership is with BMC • Bit[3:2] Reserved • Bit[4] 1: Indicates that NIOS® is requesting to MM9 ownership ID fields. • Bit[5] 1: Indicates that BMC granted MM9 ownership. • Bit[7:6] Reserved Note: PCH/CPU has read /write access to only bit[1:0] and other bits NO access. BMC has read/write access to bit[1:0] and Bit[5] and other bits NO access.
74h	Active AFM SVN info	Set by CPLD ROT; Read only for CPU/BMC

75h	Active AFM Major version	Set by CPLD ROT; Read only for CPU/BMC
76h	Active AFM Minor version	Set by CPLD ROT; Read only for CPU/BMC
77h	Recovery AFM SVN info	Set by CPLD ROT; Read only for CPU/BMC
78h	Recovery AFM Major version	Set by CPLD ROT; Read only for CPU/BMC
79h	Recovery AFM Minor version	Set by CPLD RoT; Read only for CPU/BMC
7Ah	Provisioning status 2	Status register for UFM provisioning and access commands
C0h-FFh	BMC scratchpad	Scratchpad for BMC to use, persists across DC-off/on cycles, read/write allowed from BMC, read only from CPU